

A calculator should NOT be used on the following questions.

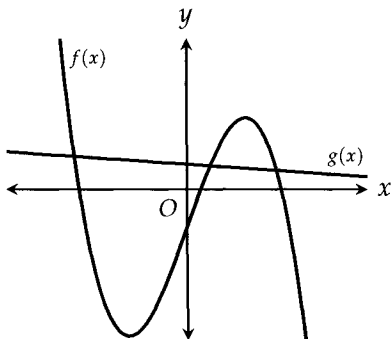
1

x	y
0	20
1	21
3	29

The table above displays several points on the graph of the function f in the xy -plane. Which of the following could be $f(x)$?

- A) $f(x) = 20x$
- B) $f(x) = x + 20$
- C) $f(x) = x - 20$
- D) $f(x) = x^2 + 20$

2

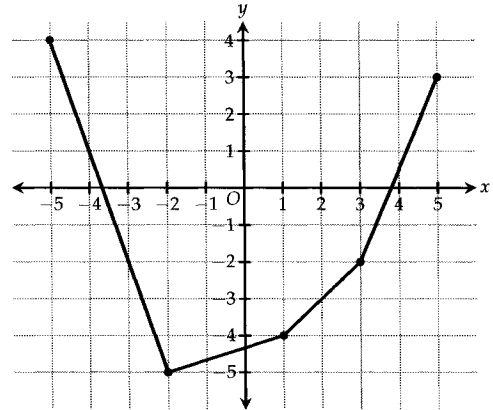


In the portion of the xy -plane shown above, for how many values of x does $f(x) = g(x)$?

- A) None
- B) One
- C) Two
- D) Three

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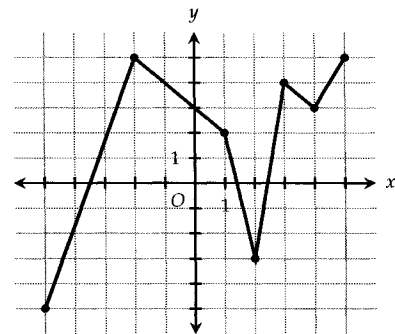
3



The graph of the function f is shown in the xy -plane above. If $f(a) = f(3)$, which of the following could be the value of a ?

- A) -4
- B) -3
- C) -2
- D) 1

4



The function f is graphed in the xy -plane above. For how many values of x does $f(x) = 3$?

- A) Two
- B) Three
- C) Four
- D) Five

5

For which of the following functions is it true that $f(-3) = f(3)$?

- A) $f(x) = \frac{2}{x}$
- B) $f(x) = \frac{x^3}{3}$
- C) $f(x) = 3x^2 + 1$
- D) $f(x) = x + 2$

6

The function f is defined by $f(x) = 3x + 2$ and the function g is defined by $g(x) = f(2x) - 1$. What is the value of $g(10)$?

- A) 61 B) 71 C) 81 D) 91

7

If $f(x) = \frac{16 + x^2}{2x}$ for all $x \neq 0$, what is the value of $f(-4)$?

- A) -8
- B) -4
- C) 4
- D) 8

8

x	0	1	2
$f(x)$	-2	3	18

Several values of the function f are given in the table above. If $f(x) = ax^2 + b$ where a and b are constants, what is the value of $f(3)$?

- A) 23
- B) 39
- C) 43
- D) 56

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If $f(x) = x^2$, for which of the following values of c is $f(c) < c$?

- A) $\frac{1}{2}$
- B) 1
- C) $\frac{3}{2}$
- D) 2

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10

If the graph of the function f has x -intercepts at -3 and 2, and a y -intercept at 12, which of the following could define f ?

- A) $f(x) = (x + 3)^2(x - 2)$
- B) $f(x) = (x + 3)(x - 2)^2$
- C) $f(x) = (x - 3)^2(x + 2)$
- D) $f(x) = (x - 3)(x + 2)^2$

11

$$f(x) = x^2 + 1$$

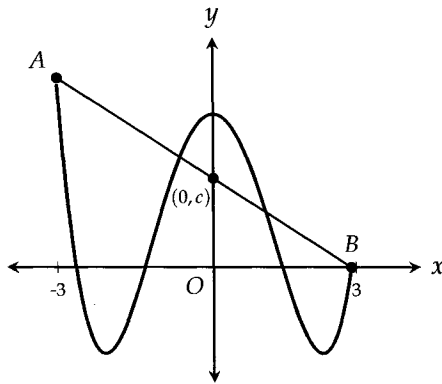
$$g(x) = x^2 - 1$$

The functions f and g are defined above. What is the value of $f(g(2))$?

- A) 3
- B) 5
- C) 10
- D) 17

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12



The graph of the function f and line segment $\overline{AB}$ are shown in the xy -plane above. For how many values of x between -3 and 3 does $f(x) = c$?

- A) 3 B) 4 C) 5 D) 6

13

x	$f(x)$
-4	3
-2	5
0	2
2	16
3	4
4	8

The table above gives some values for the function f . If $g(x) = 2f(x)$, what is the value of k if $g(k) = 8$?

- A) 2
B) 3
C) 4
D) 8

14

$$f(x) = \sqrt{x-2}$$

The function f is defined above for all $x \geq 2$. Which of the following is equal to $f(18) - f(11)$?

- A) $f(3)$
B) $f(5)$
C) $f(6)$
D) $f(7)$

A calculator is allowed on the following questions.

15

$$y = \frac{x+1}{x-1}$$

Which of the following points in the xy -plane is NOT on the graph of y ?

- A) $(-2, \frac{1}{3})$
B) $(-1, 0)$
C) $(0, -1)$
D) $(1, 2)$

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Let the function g be defined by $g(x) = \sqrt{3x}$. If $g(a) = 6$, what is the value of a ?

- A) 3
B) 6
C) 9
D) 12

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Questions 17-18 refer to the following information.

x	$f(x)$	$g(x)$
-2	3	4
-1	5	2
0	-2	-3
1	3	5
2	6	7
3	7	1

The functions f and g are defined for the six values of x shown in the table above.

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What is the value of $f(g(-1))$?

- A) -2
- B) 3
- C) 5
- D) 6

18

If $g(c) = 5$, what is the value of $f(c)$?

- A) -2
- B) 3
- C) 5
- D) 6

19

If $f(x) = -3x + 5$ and $\frac{1}{2}f(a) = 10$, what is the value of a ?

- A) -8
- B) -5
- C) 5
- D) 8

20

x	$f(x)$
0	-4
1	-8
2	3
3	6
4	7
5	2
6	4
7	5

Several values of the function f are given in the table above. If the function g is defined by $g(x) = f(2x - 1)$, what is the value of $g(3)$?

- A) 2
- B) 6
- C) 5
- D) 7

21

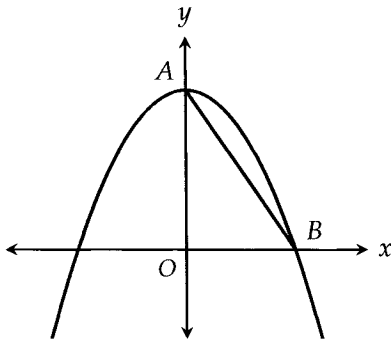
$$f(x) = 4x - 3$$

$$g(x) = 3x + 5$$

The functions f and g are defined above. Which of the following is equal to $f(8)$?

- A) $g(1)$
- B) $g(3)$
- C) $g(5)$
- D) $g(8)$

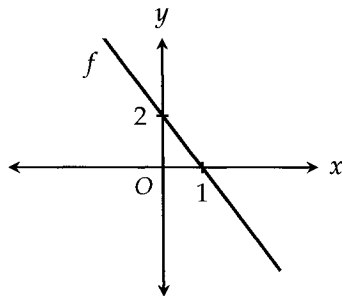
22



The graph of the function $y = 9 - x^2$ is shown in the xy -plane above. What is the length of $\overline{AB}$?

- A) $3\sqrt{2}$
- B) $3\sqrt{10}$
- C) 9
- D) $9\sqrt{10}$

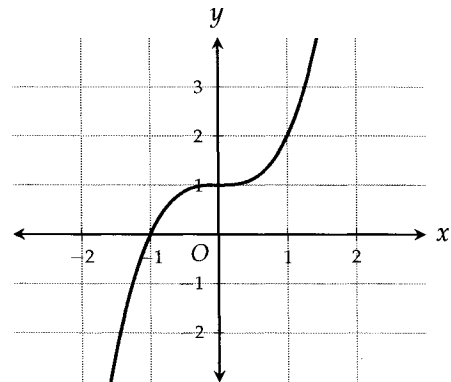
23



The function f is graphed in the xy -plane above. If the function g is defined by $g(x) = f(x) + 4$, what is the x -intercept of $g(x)$?

- A) -3
- B) -1
- C) 3
- D) 4

24



The function $f(x) = x^3 + 1$ is graphed in the xy -plane above. If the function g is defined by $g(x) = x + k$, where k is a constant, and $f(x) = g(x)$ has 3 solutions, which of the following could be the value of k ?

- A) -1
- B) 0
- C) 1
- D) 2

25

In the xy -plane, the function $y = ax + 12$, where a is a constant, passes through the point $(-a, a)$. If $a > 0$, what is the value of a ?

- A) 3 B) 4 C) 5 D) 6